Transcutaneous Spinal Stimulation & Gait Training Enhances Spinal Motor Network Excitability in Individuals with TBI

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***Abstract*— Transcutaneous spinal cord stimulation (tSCS) combined with task specific gait training may enhance spinal motor network excitability following traumatic brain injury (TBI). This study evaluated changes in spinal motor network excitability and functional mobility in a 20 year old female with chronic TBI (23 months post injury) following a 6 week treadmill based gait training with tSCS. Spinal motor evoked potentials (sMEPs) were assessed under modulated and unmodulated stimulation conditions. Resting motor threshold (RMT), recruitment curves, and sigmoidal model parameters (I50) were analyzed. Recruitment curve modeling was performed using sigmoidal fits to quantify changes in motor pool activation. RMT decreased over time and recruitment curves demonstrated a leftward shift with reduced stimulation intensity required to elicit motor responses. Modulated stimulation produced lower thresholds and more consistent recruitment compared to unmodulated stimulation. Functional mobility improved, with Timed Up and Go (TUG) decreasing from 10.62 s at baseline to 8.21 s at long term follow up. These findings suggest that tSCS combined with gait training may enhance spinal motor network excitability and contribute to functional improvements in individuals with chronic TBI.**

I. Introduction

Traumatic brain injury (TBI) is a leading cause of long-term disability that affects 2.8 million people annually in the US [1]. TBI can disrupt sensorimotor control, resulting in gait and balance impairments that may persist during the chronic stages of recovery. This includes slower walking speed, decreased step length, postural instability, and impaired weight transfer [2]. All of these pathological changes interfere with higher-level mobility skills necessary for full return to activities of daily living, full community participation, and vocational activities [3]. Activity-dependent neurorehabilitation remains the primary intervention for promoting long-term motor recovery following TBI [3]. This approach is based on the principle of neuroplasticity, whereby repetitive, task-specific, goal-directed motor practice promotes reorganization of neural circuits and facilitates recovery of motor function [4]. Recent advances in rehabilitation have evaluated the combination of rehabilitative training with neuromodulation of the brain and/or the spinal cord as means to enhance the excitability of motor circuits and to increase training efficacy promoting motor recovery [5,6]. The proposed investigation will evaluate the efficacy and safety of transcutaneous spinal stimulation (tSCS) to promote functional motor recovery for individuals with chronic TBI without eliciting maladaptive changes. The fundamental hypothesis guiding this proposal is that transcutaneous spinal cord stimulation (tSCS) combined with gait training (WALK) for individuals with chronic TBI will lead to improved modulation of spinal motor network excitability.

II. Methods

A. Participants

A single participant (20 year old female) diagnosed with TBI (23 months post injury, height 1.63 m, 54.43 kg) enrolled in the study. All procedures were conducted in accordance with the institutional review board and informed consent was obtained prior to participation.

Participants completed a 6 week (3 times per week, 18 sessions) treadmill based gait training program (WALK) focused on improving symmetry, kinematic quality and walking intensity. Transcutaneous spinal cord stimulation (tSCS) was delivered using a portable stimulator (Digitimer DS8R®, USA). Biphasic stimulation was applied over the thoracolumbar region using a cathode electrode (7.5 X 13cm) positioned between T11/T12 and L1/L2 vertebrae (corresponds to spinal networks that innervate leg muscles) and two anodes placed bilaterally over the iliac crests (as shown in Fig. 1). During treadmill based gait training stimulation was delivered at 30 Hz base frequency, a 5 kHz carrier

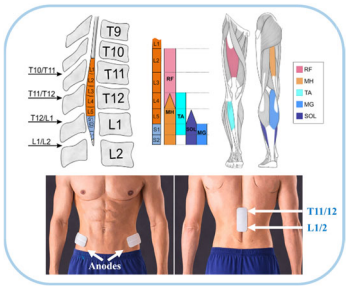

Fig. 1. tSCS electrode placement.

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frequency and a 100 µs pulse width.

B. Neurophysiological Assessment

Spinal motor evoked potential (sMEPs) were recorded from bilateral lower extremity muscles while the participant was supine: vastus lateralis (VL), rectus femoris (RF), medial hamstring (MH), tibialis anterior (TA), and medial gastrocnemius (MG). Stimulation intensity was increased from 10-250 mA in 10 mA increments. At each intensity, 5 stimuli were delivered (5 second interstimulus interval). Assessments were performed under two conditions, modulated (5kHz carrier; 100 µs pulse width) and unmodulated (no carrier; 1000 µs pulse width).

- *Resting Motor Threshold (RMT)* – the lowest stimulation intensity at which ≥ 3 out of 5 responses exceeded 0.05 mV in any recorded muscle.
- *Recruitment Curve Analysis* – peak to peak sMEP amplitudes were plotted as a function of stimulation intensity. Sigmoidal (logistic) functions were fit to recruitment curves to extract I50 (the stimulation intensity corresponding to 50% of maximal response); slope (gain) and plateau amplitude. Both lowest threshold muscle and representative muscle (LMG) analyses were performed.

III. Results

Transcutaneous spinal cord stimulation (tSCS) combined with gait training resulted in measurable changes in spinal excitability and recruitment characteristics. The participant completed all intervention sessions and stimulation was tolerated during treadmill gait training sessions with no adverse events. Improvements in excitability corresponded with observed improvements in gait performance (Fig. 2).

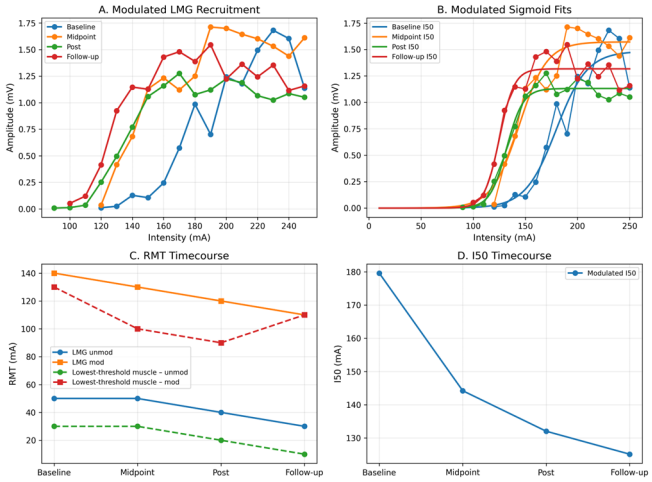

Fig. 2. (A) Modulated LMG recruitment curves show a leftward shift across timepoints. (B) Sigmoidal fits demonstrate reduced I50, indicating increased excitability. (C) RMT decreases over time for LMG and the lowest-threshold muscle. (D) I50 time course shows reduced stimulation required to reach 50% maximal response.

RMT decreased over time, indicating increased excitability of spinal motor networks (Fig. 2C). The lowest threshold muscle demonstrated a progressive reduction in RMT across sessions, with the largest decrease observed at follow up. Left medial gastrocnemius (LMG) RMT showed a similar but less pronounced trend. Recruitment curves demonstrated a leftward shift across time points under the modulated condition (Fig. 2A), indicating reduced stimulation intensity required to elicit motor responses. This shift was accompanied by increased motor response amplitudes at lower intensities. Sigmoid modeling revealed reduced I50 values over time (modulated condition) (Fig. 2B), increased recruitment efficiency and earlier saturation of motor responses.

These findings indicate enhanced responsiveness of spinal motor circuits following the intervention. Modulated stimulation produced lower thresholds, greater response amplitudes and more consistent recruitment patterns compared to the unmodulated stimulation (Fig. 2D).

Functional mobility improved following the treadmill gait training and tSCS intervention. The Timed Up and Go (TUG) performance improved from 10.62 at baseline to 8.21 at long term follow up indicating meaningful gains in mobility and functional performance.

IV. Discussion

Preliminary data demonstrates that tSCS combined with gait training may enhance spinal motor network excitability in a chronic TBI participant. These neurophysiological changes were accompanied by improvements in functional mobility suggesting that improved spinal excitability translated to improved motor performance. Reductions in RMT and leftward shifts in recruitment curves suggest improved recruitment and increased responsiveness of spinal circuitry. These findings are consistent with the neuromodulatory effects of tSCS, which may enhance afferent input and facilitate transmission of residual descending commands. The differences between lowest muscle and LMG specific analysis highlights that global excitability metrics may not fully capture muscle specific adaptations. Utilization of sigmoid modeling may provide additional information about motor system behavior. Preliminary data indicates that modulated stimulation demonstrated larger effects compared to unmodulated stimulation. This may begin to support the role of high frequency carrier stimulation in reducing impedance and enhancing neural recruitment.

More research is needed in a larger sample with individuals with varying levels of impairment to fully understand how tSCS combined with gait training can impact overall recovery. These neurophysiological changes suggest that tSCS combined with gait training may promote meaningful reorganization of corticospinal activity connectivity.

V. Conclusion

This study begins to demonstrate that tSCS combined with treadmill based gait training may enhance spinal motor network excitability and improve functional mobility in a chronic TBI participant. Reductions in RMT and shifts in recruitment characteristics indicate increased responsiveness of spinal motor circuits. Future research will expand this preliminary data and continue to understand the potential of tSCS combined with gait training in chronic TBI.

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